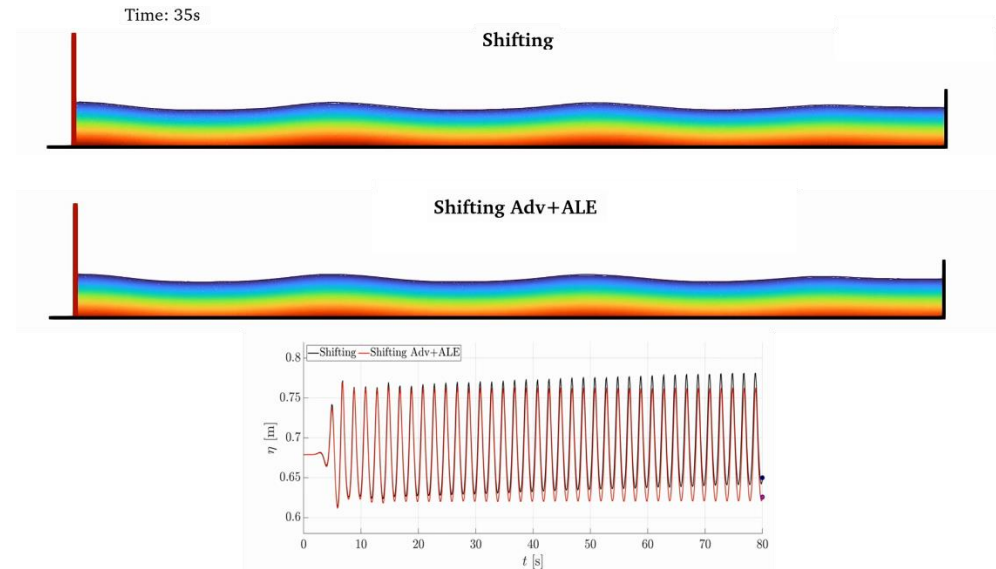


01_WAVESDAMPING

- 2-D regular waves are generated and propagated in a numerical wave flume. Two simulations obtained using Shifting and Shifting advanced +ALE are provided.
The inclusion of Arbitrary Lagrangian_Eulerian terms with the novel Shifting advanced method improve the volume conservation.

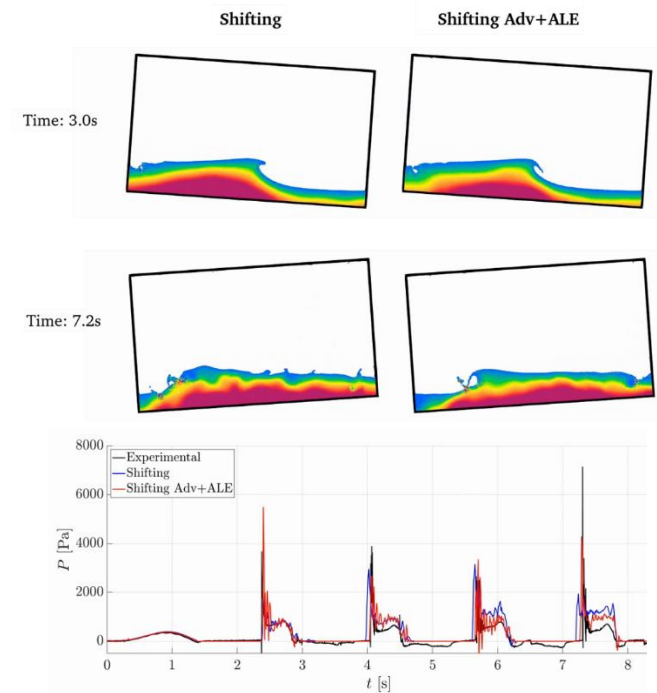
[Video](#)



02_SLOSHING

- The case will reproduce a sloshing experiment that includes moving boundaries. The proposed test case is the SPHERIC Benchmark Test Case #10. The numerical pressures can be obtained using Shifting and Shifting advanced + ALE and compared with the experimental values detected at the pressure sensor during the experiment.

[Video](#)

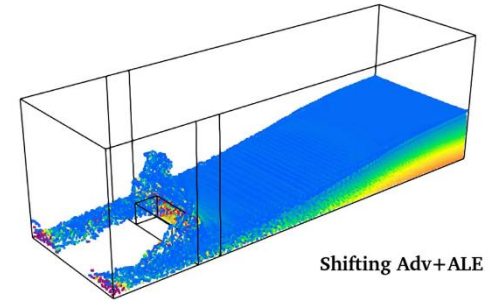
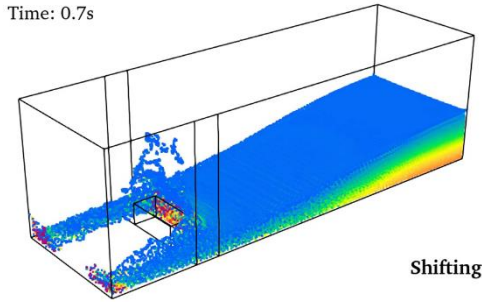


03_DAMBREAK

- The case will reproduce the dam break experiment that includes moving boundaries proposed as SPHERIC Benchmark Test Case #2. The numerical pressures can be obtained using Shifting and Shifting advanced + ALE and compared with the experimental values detected at the pressure sensor during the experiment.

[Video](#)

Time: 0.7s



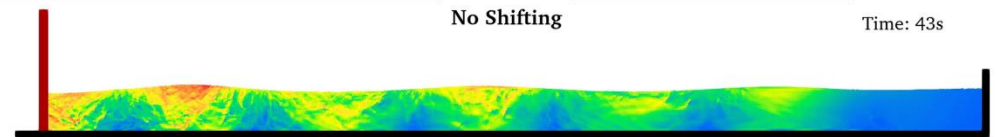
04_WAVESDAMPINGNC

- 2-D regular waves are generated and propagated in a numerical wave flume. Two simulations obtained using Shifting and Shifting advanced +ALE+ Non-conservative pressure formulation are provided. The inclusion of the non-conservative pressure formulation reduces the numerical dissipation in case of small smoothing length to particles size ratio

[Video](#)

No Shifting

Time: 43s



Shifting Adv+ALE+NCPress

